

FIMsim Test Cases

Two ready-to-use study areas so you can try FIMsim end to end without any input data of your own. Each test case walks through one complete flood-model setup:

Test case	AOI shapefile	Location	Model	Flood event
1	A0I_1_Neuse/A0I_1.shp	Neuse River, North Carolina	LISFLOOD-FP	Hurricane Matthew — October 2016
2	A0I_2_Texas/A0I_2.shp	Village Creek, Texas	TRITON	Hurricane Harvey — August 2017

What you need: FIMsim (web app or desktop installer — see the main README on GitHub) and an internet connection. You provide the AOI shapefile plus the start and end dates of the flood event you want to simulate — all terrain, land-cover, river-network, and discharge data are downloaded automatically.

Test Case 1 — LISFLOOD-FP · Neuse River, North Carolina

AOI: A0I_1_Neuse/A0I_1.shp · 293.4 km² · CRS EPSG:26917 (NAD83 / UTM 17N) · Event: Hurricane Matthew, 2016-10-05 → 2016-10-20

In early October 2016, Hurricane Matthew brought extreme rainfall to eastern North Carolina, producing record flooding along the Neuse River. This walkthrough builds the complete LISFLOOD-FP input package for a 15-day simulation of that event, using the USGS stream gage inside the AOI for the inflow hydrograph.

Step 1 — Project

Open FIMsim and choose the **LISFLOOD-FP** model. Create a new project — pick any project name and an empty folder where all outputs will be written.

FIMsim | Flood Inundation Model Simulation Tool v1.0 (LISFLOOD-FP)

1. Project 2. AOI 3. DEM 4. Manning 5. BCI 6. BDY 7. PAR File

☒ New project ☐ Open existing project

Base directory: /pnikrou/Documents [Browse...](#)

Project name: LISFLOOD_project

[Run](#)

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Step 2 — AOI

Click **Browse** and select `AOI_1.shp`. You can load several AOI files, or pick one or more features from a shapefile that contains multiple polygons — for this test case it is a single feature: tick it and click **Add to confirmed AOIs**.

The panel gives an overview of the selected AOI: geographic location, area, HUC6/HUC8 codes, CRS, the main river, and any USGS gages found inside the AOI.

Expected: - Area **293.40 km²**, State **North Carolina (NC)**, CRS **EPSG:26917** - HUC6 **030202** | HUC8 **03020201, 03020202, 03020203** - Main river: **Neuse River** - USGS gage found: **02089000 — Neuse River near Goldsboro, NC**

FIMsim | Flood Inundation Model Simulation Tool v1.0 (LISFLOOD-FP)
1. Project
2. AOI
3. DEM
4. Manning
5. BCI
6. BDY
7. PAR File

★ **Select your Area(s) of Interest (AOI).** Browse for a shapefile or GeoPackage and pick the feature(s) you want. To add more AOIs from another file, click " + Add another AOI file". When you're done, click "Add to confirmed AOIs" — then click "Next step ►" at the bottom to continue.

AOI file:
/Users/pnikrou/Documents/Code_Chapter2/lisflood_prep_app/AOIs/AOI_1.shp
Browse...
+ Add another AOI file

Use	#	Name	Area (km²)	State	Centroid (lon, lat)
☒	1	AOI_1	293.40	NC	-78.023, 35.370

Select all features
Deselect all

1 AOI feature(s) ready to confirm from 1 file(s).

Add to confirmed AOIs

Selected AOI details

AOI_1 (from AOI_1.shp)
Area: 293.40 km²
CRS: NAD83 / UTM zone 17N (EPSG:26917)
State: North Carolina (NC)
HUC6: 030202 | **HUC8:** 03020201, 03020202, 03020203
Main river: Neuse River
USGS gages in AOI: 1 found:
02089000 NEUSE RIVER NEAR GOLDSBORO, NC

Map preview

United States — North Carolina

North Carolina — AOI location

Neuse River

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Step 3 — DEM

Keep **Download from 3DEP (USGS)** and set the DEM cell size to **10 m**. Click **Run**. FIMsim downloads the elevation tiles, mosaics, resamples, reprojects to the AOI's UTM zone, and clips to the AOI.

Expected: a 1630 × 1800 px DEM at 10 m resolution (DEM_AOI_1.tif), previewed with elevations of roughly 15–65 m, plus dem.ascii in lisflood-files/ .

FIMsim | Flood Inundation Model Simulation Tool v1.0 (LISFLOOD-FP)

1. Project

2. AOI

3. DEM

4. Manning

5. BCI

6. BDY

7. PAR File

1 AOI confirmed.

3. Digital Elevation Model (DEM)

DEM source:

☒ Download from 3DEP (USGS)

☐ I have a DEM raster

DEM cell size: 10.0 m

Run

All 1 AOI(s) processed.

Per-AOI DEM outputs — click an AOI to preview its DEM

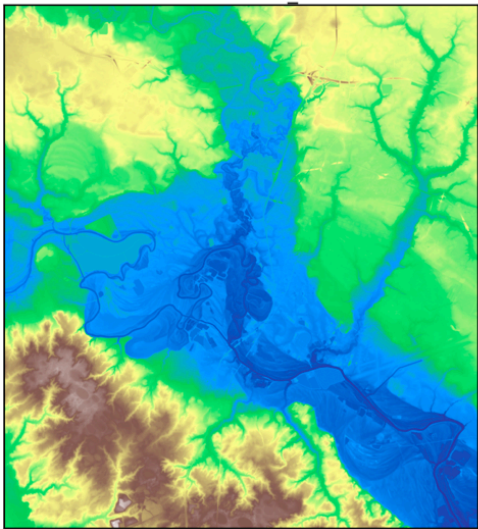
AOI_1

DEM preview

DEM Information

AOI	AOI_1
CRS	EPSG:26917
Width x Height	1630 x 1800 px
Resolution	10.00 x 10.00 m
Bounds (W, S)	762300.000, 3909100.000
Bounds (E, N)	778600.000, 3927100.000
File	DEM_AOI_1.tif

DEM — AOI 1



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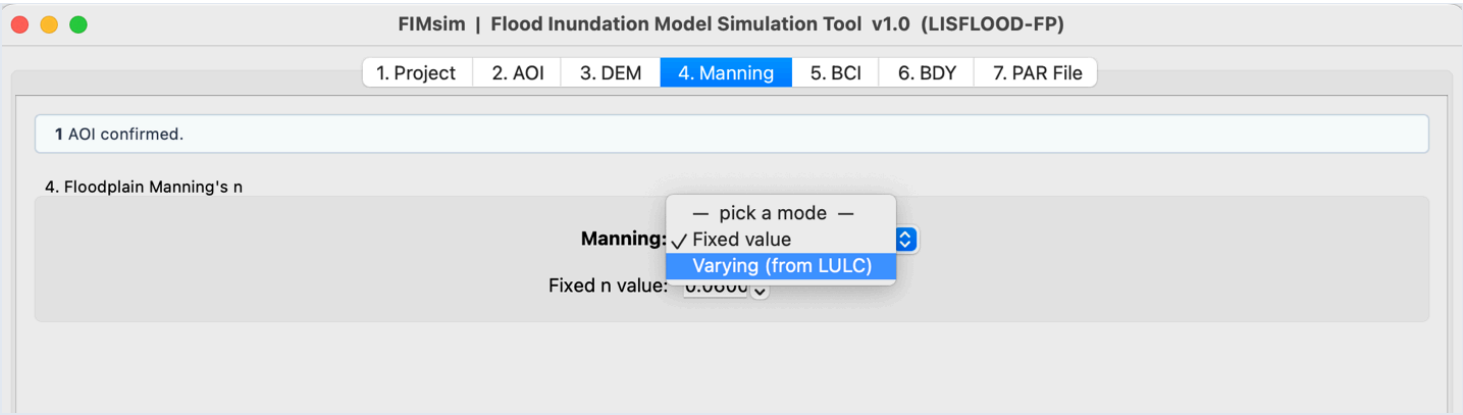
Clear log

About

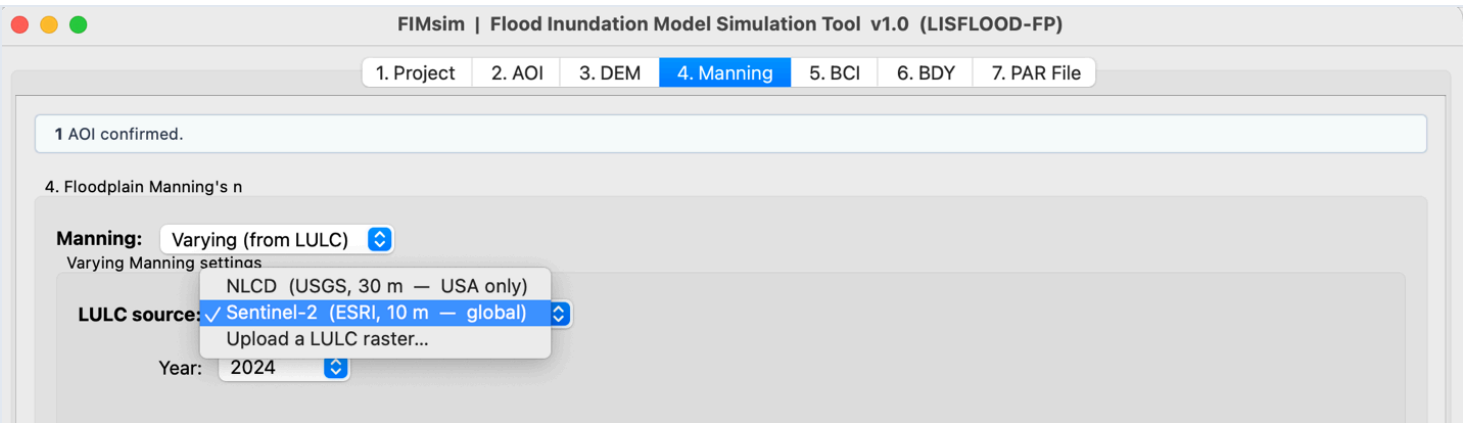
Help

Step 4 — Manning

First choose between **Fixed** roughness (one value everywhere) and **Varying** roughness (derived from land cover):



Select **Varying**, then pick the LULC source and year — for this test case use **NLCD**. FIMsim downloads the land-cover raster, resamples/reprojects/clips it to the DEM grid, and converts it to a Manning's n map using the editable lookup table:



Expected: side-by-side LULC and Manning's n previews with per-class percentages (the AOI is dominated by cultivated crops and woody wetlands). Files produced: `LULC_AOI_1_<year>.tif`, `ManningN_AOI_1.tif`, and `lulc.ascii`.

FIMsim | Flood Inundation Model Simulation Tool v1.0 (LISFLOOD-FP)

1. Project

2. AOI

3. DEM

4. Manning

5. BCI

6. BDY

7. PAR File

Manning: Varying (from LULC)

Varying Manning settings

LULC source: Sentinel-2 (ESRI, 10 m — global)

Year: 2024

LULC class → Manning n mapping (Min/Max are reference bounds; Avg is editable and clamped):

Code	Land Cover Class	Min n	Max n	Avg n (editable)
6	Scrub/Shrub	0.050	0.090	0.0700
7	Built Area	0.080	0.140	0.1100
8	Bare Ground	0.025	0.045	0.0350
9	Snow/Ice	0.020	0.050	0.0350
10	Clouds	—	—	0.1050
11	Rangeland	0.030	0.060	0.0450

Prepare Manning File

All 1 AOI(s) processed.

Per-AOI Manning outputs — click an AOI to preview its Manning map

AOI_1

LULC Manning preview

Land Cover Breakdown

Code	Type	Area (km²)	% area
7	Built ...	104.789	35.7%
2	Trees	99.673	34.0%
5	Crops	49.949	17.0%
11	Range...	27.197	9.3%
1	Water	11.085	3.8%
8	Bare ...	0.618	0.2%
4	Flood...	0.089	0.0%
Total		293.399	100.0%

LULC Map

LULC — AOI 1

Manning's n Map

Manning n — AOI 1

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Clear log

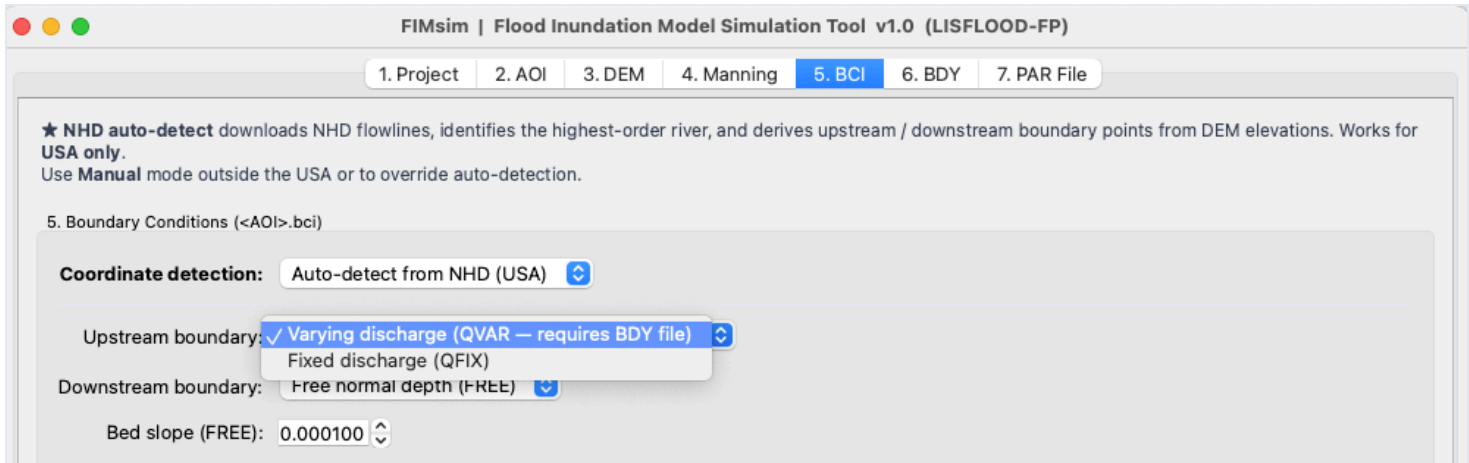
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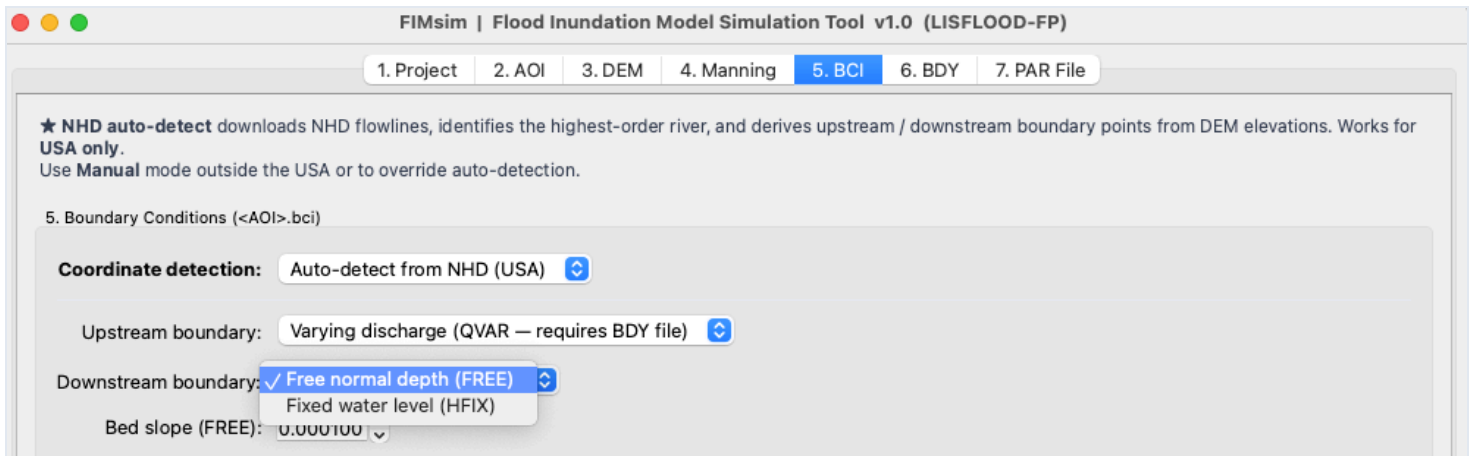
Step 5 — BCI (Boundary Conditions)

Keep Auto-detect from NHD (USA).

The upstream boundary can be a **Varying discharge (QVAR)** — driven by the hydrograph from the next step — or a **Fixed discharge (QFIX)**. Select **QVAR**:



The downstream boundary can be a **Free normal depth (FREE)** or a **Fixed water level (HFIX)**. Select **FREE** with bed slope **0.0001** :



Click **Write .bci file(s)**. FIMsim downloads the NHD river network, identifies the main river, and derives the upstream/downstream boundary points from DEM elevations.

Expected: the BCI preview shows the Neuse River crossing the domain with the upstream point (orange) on the west edge and the downstream point (red) on the southeast corner, and the generated `AOI_1.bci` contains a `P ... QVAR upstream1` line and an `S ... FREE 0.0001` line.

FIMsim | Flood Inundation Model Simulation Tool v1.0 (LISFLOOD-FP)

1. Project

2. AOI

3. DEM

4. Manning

5. BCI

6. BDY

7. PAR File

★ **NHD auto-detect** downloads NHD flowlines, identifies the highest-order river, and derives upstream / downstream boundary points from DEM elevations. Works for **USA only**.
Use **Manual** mode outside the USA or to override auto-detection.

1 AOI confirmed.

5. Boundary Conditions (<AOI>.bci)

Coordinate detection: Auto-detect from NHD (USA)

Upstream boundary: Varying discharge (QVAR — requires BDY file)

Downstream boundary: Free normal depth (FREE)

Bed slope (FREE): 0.000100

Write .bci file(s)

All 1 AOI(s) processed.

Per-AOI BCI outputs — click an AOI to preview its BCI on the map

AOI_1

BCI preview

P	762399.395	3917210.126	QVAR
upstream1			
S	777016.365	778600.000	FREE
0.0001			

BCI — AOI_1

Main river

Upstream

Downstream

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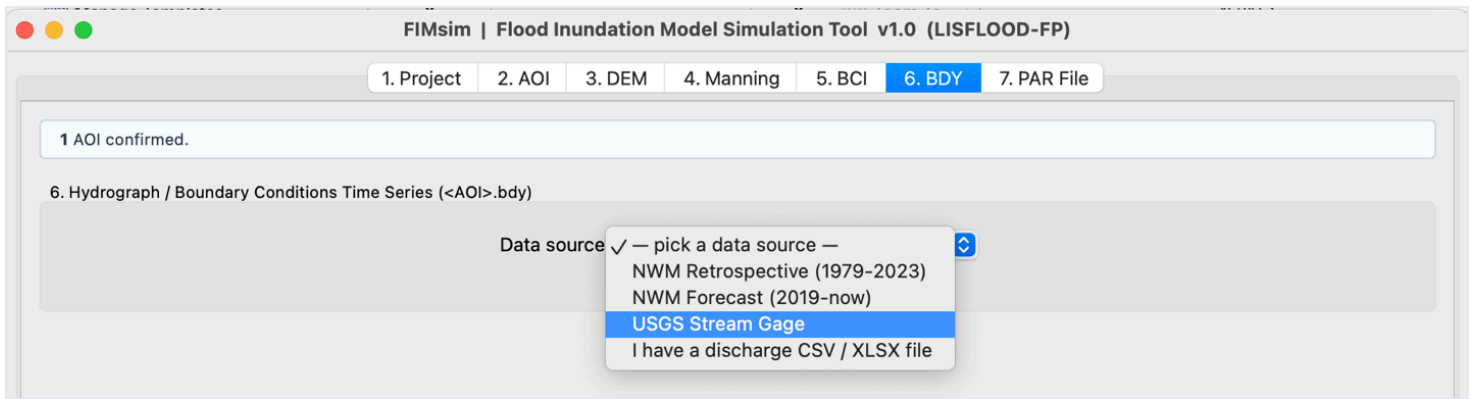
Clear log

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Step 6 — BDY (Hydrograph)

The hydrograph step offers several data sources: **NWM Retrospective** (1979–2023), **NWM Forecast** (2019–present; short/medium/long range), **USGS Stream Gage**, or your own CSV file:



For this test case select **USGS Stream Gage** with the gage found in Step 2:

- **Gage number:** 02089000
- **Event start:** 2016-10-05 00:00
- **Event end:** 2016-10-20 00:00
- **Time interval:** 1.00 hours

Click **Create BDY File**.

Expected: the hydrograph preview shows the Hurricane Matthew flood wave — rising sharply on **October 9** and peaking near **1,500 m³/s around October 12–13** — and `AOI_1.bdy` is written.

1. Project
2. AOI
3. DEM
4. Manning
5. BCI
6. BDY
7. PAR File

1 AOI confirmed.

6. Hydrograph / Boundary Conditions Time Series (<AOI>.bdy)

Data source: USGS Stream Gage
Gage number: 02089000
★ Data available at waterdata.usgs.gov | 15-min readings resampled to chosen interval
Event start: 2016-10-05 00:00
Event end: 2016-10-20 00:00
Time interval: 1.00 hours

Create BDY File

All 1 AOI(s) processed.

Per-AOI BDY outputs — click an AOI to preview its hydrograph

AOI_1

USGS

Hydrograph preview

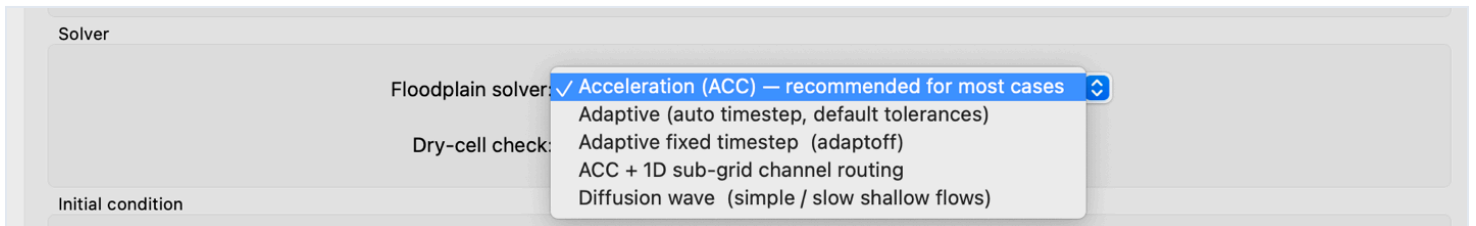
Hydrograph — AOI_1

Time (Date)	Discharge (m³/s)
2016-10-05	~100
2016-10-07	~100
2016-10-09	~500
2016-10-11	~1000
2016-10-13	~1400
2016-10-15	~800
2016-10-17	~400
2016-10-19	~200

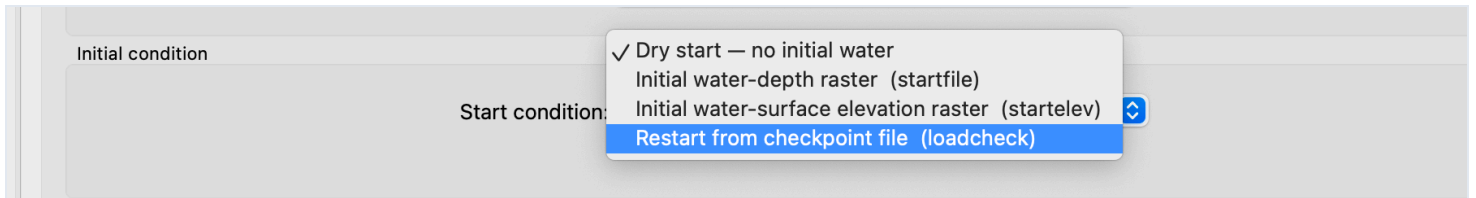
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Step 7 — PAR (Parameter file)

The final step assembles the LISFLOOD-FP parameter file. You can choose between solvers — the **Acceleration (ACC)** solver is recommended for most cases:



...and set the initial condition (keep **Dry start — no initial water**):



The simulation time is derived automatically from the hydrograph. Keep the remaining defaults (1 s initial timestep, 3600 s save interval) — there is also an *Extra PAR keywords* box for any additional LISFLOOD-FP keywords you may want:

1. Project
2. AOI
3. DEM
4. Manning
5. BCI
6. BDY
7. PAR File

1 AOI confirmed.

7. PAR file

Output file names

PAR filename: AOI_1.par

Result prefix (resroot): AOI_1

Results folder: AOI_1_Results

Simulation timing

Simulation time: 1281600.00 s

Initial timestep: 1.000 s

Save interval: 3600.00 s

Mass-balance interval: 3600.00 s

Solver

Floodplain solver: Acceleration (ACC) — recommended for most cases

Dry-cell check: leave_default_off (LISFLOOD default)

Initial condition

Start condition: Dry start — no initial water

Checkpointing restart

☐ Enable checkpointing every 1.00 computation-hours

Output options

☒ elevoff — suppress water-surface elevation grids
☐ depthoff — suppress water-depth grids
☐ binary_out — write rasters as binary instead of ASCII
☐ hazard — write hazard (depth × velocity) grids
☐ mint_hk — record max depth/velocity at each massint interval
☐ qoutput — write boundary-discharge timeseries to file
☐ sgc_enable — activate sub-grid channel scheme (SGC)
☐ overpass — save snapshot at time 100000.00 s

Extra PAR keywords

One keyword (or keyword value) per line
e.g. cuda
SGCwidth 5.0

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Expected: AOI_1.par is written. The lisflood-files/ folder now contains the complete LISFLOOD-FP input package — dem.ascii, lulc.ascii, AOI_1.bci, AOI_1.bdy, AOI_1.par — ready to run with any LISFLOOD-FP executable.

Test Case 2 — TRITON · Village Creek, Texas (Hurricane Harvey)

AOI: `A0I_2_Texas/A0I_2.shp` · 390.1 km² · CRS EPSG:26914 (NAD83 / UTM 14N) · Event: Hurricane Harvey, 2017-08-24 → 2017-09-10

In late August 2017, Hurricane Harvey stalled over southeast Texas and produced the heaviest tropical rainfall ever recorded in the United States. Village Creek — a tributary of the Neches River near Kountze, TX — experienced record flooding. This walkthrough builds the complete TRITON input package for that event using the NWM Retrospective discharge for the automatically detected reach.

Step 1 — Project

Open FIMsim and choose the **TRITON** model. Create a new project — pick any project name and an empty output folder.

The screenshot shows the '1. Project' tab of the FIMsim v1.0 (TRITON) application. The window has a title bar with standard macOS window controls (red, yellow, green buttons) and the text 'FIMsim | Flood Inundation Model Simulation Tool v1.0 (TRITON)'. Below the title bar is a horizontal tab bar with seven tabs: '1. Project' (active, highlighted in blue), '2. AOI', '3. DEM', '4. Friction', '5. BC', '6. Hydrograph', and '7. Config'. The main content area of the '1. Project' tab contains the following elements:

- Two radio buttons at the top left: 'New project' (selected) and 'Open existing project'.
- A 'Base directory:' label followed by a text input field containing '/pnikrou/Documents' and a 'Browse...' button to its right.
- A 'Project name:' label followed by a text input field containing 'TRITON_Project'.
- A large blue 'Run' button on the left side of the main area.

At the bottom of the window is a horizontal bar with six buttons: 'Back to main page', 'Previous step', 'Next step', 'Clear log', 'About', and 'Help'.

Step 2 – AOI

Click **Browse** and select `AOI_2.shp`, tick the single feature, and click **Add to confirmed AOIs**.

Expected: - Area **390.13 km²**, State **Texas (TX)**, CRS **EPSG:26914** - HUC6 **120200** | HUC8 12020003, 12020006, 12020007 - Main river: **Village Creek** - USGS gage found: **08041500** — **Village Ck nr Kountze, TX**

FIMsim | Flood Inundation Model Simulation Tool v1.0 (TRITON)

1. Project

2. AOI

3. DEM

4. Friction

5. BC

6. Hydrograph

7. Config

★ **Select your Area(s) of Interest (AOI).** Browse for a shapefile or GeoPackage and pick the feature(s) you want. To add more AOIs from another file, click " + Add another AOI file". When you're done, click "Add to confirmed AOIs" — then click "Next step ►" at the bottom to continue.

+ Add more AOIs

Confirmed AOIs — click a row to see details below

1. AOI_02shp (TX)

Remove

Selected AOI details

AOI_02shp (from AOI_02shp.shp)

Area: 390.13 km²

CRS: NAD83 / UTM zone 14N (EPSG:26914)

State: Texas (TX)

HUC6: 120200 | HUC8: 12020003, 12020006, 12020007

Main river: Village Creek

USGS gages in AOI: 1 found:

08041500 Village Ck nr Kountze, TX

Map preview

United States — Texas

Texas — AOI location

Village Creek

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Step 3 — DEM

Keep **Download from 3DEP (USGS)** with a **10 m** cell size and click **Run**.

Expected: a 1400 × 2800 px DEM at 10 m resolution (`DEM_<AOI>.tif`), with elevations of roughly 3–38 m and Village Creek clearly visible as the low (blue) corridor, plus `dem.asc` in `triton-files/`.

FIMsim | Flood Inundation Model Simulation Tool v1.0 (TRITON)

1. Project2. AOI3. DEM4. Friction5. BC6. Hydrograph7. Config

DEM source:

Download from 3DEP (USGS)

I have a DEM raster

DEM cell size: 10.0 m

Run

All 1 AOI(s) processed.
Per-AOI DEM outputs — click an AOI to preview its DEM

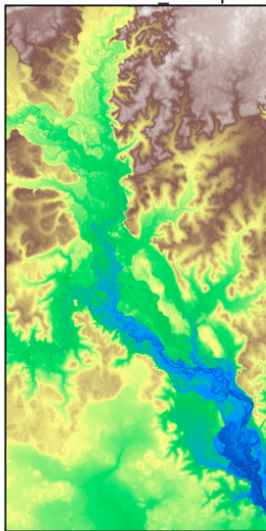
AOI_02shp

DEM preview

DEM Information

AOI	AOI_02shp
CRS	EPSG:26914
Width x Height	1400 x 2800 px
Resolution	10.00 x 10.00 m
Bounds (W, S)	949500.000, 3357500.000
Bounds (E, N)	963500.000, 3385500.000
File	DEM_AOI_02shp.tif

DEM — AOI_02shp



Elevation (m)

3530252015105

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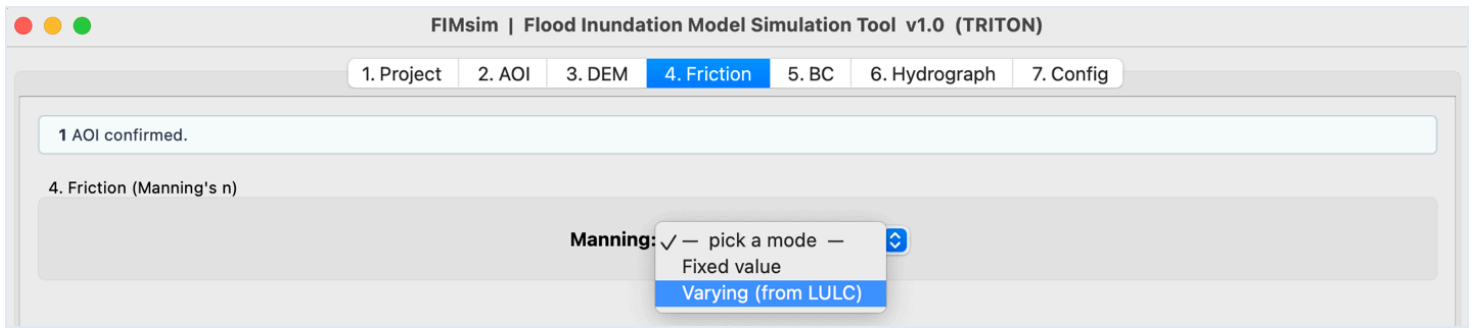
Clear log

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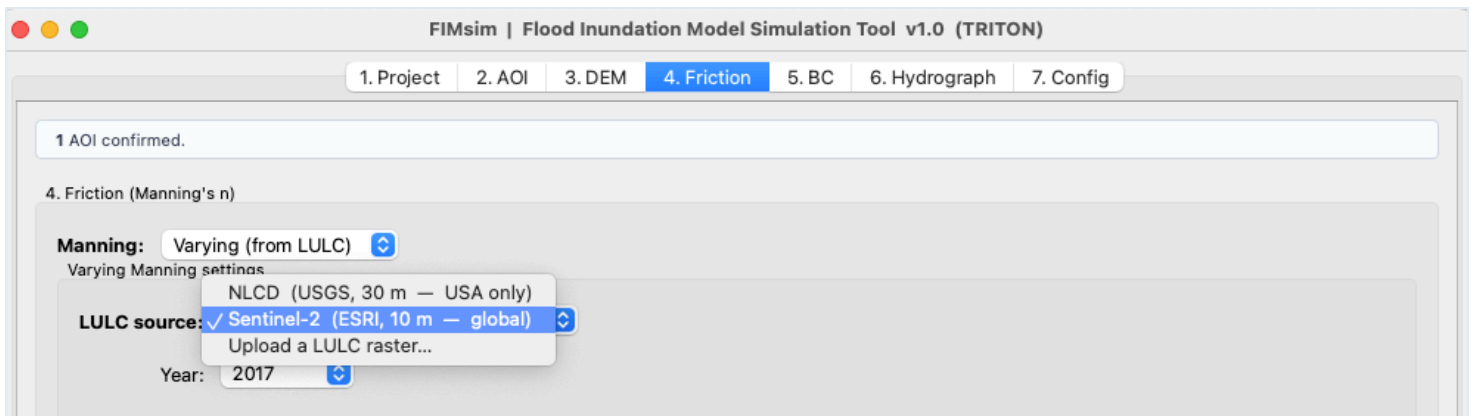
Help

Step 4 — Friction

First choose between **Fixed** and **Varying** roughness:



Select **Varying**, then pick the LULC source and year. In Test Case 1 we used NLCD; this time select **Sentinel-2 (ESRI, 10 m — global)** with year **2017** (the year of the event) — demonstrating the globally available land-cover option:



Expected: LULC and Manning's n previews (the AOI is dominated by forest and woody wetlands). Files produced: `LULC_<AOI>_<year>.tif`, `ManningN_<AOI>.tif`, and the TRITON friction raster `friction.asc` snapped to the DEM grid.

FIMsim | Flood Inundation Model Simulation Tool v1.0 (TRITON)

1. Project

2. AOI

3. DEM

4. Friction

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7. Config

LULC source: Sentinel-2 (ESRI, 10 m — global)

Year: 2017

LULC class → Manning n mapping (Min/Max are reference bounds; Avg is editable and clamped):

Code	Land Cover Class	Min n	Max n	Avg n (editable)
1	Water	0.025	0.035	0.0300
2	Trees	0.080	0.140	0.1100
3	Grass	0.030	0.060	0.0450
4	Flooded Vegetation	0.060	0.120	0.0900
5	Crops	0.025	0.055	0.0400
6	Scrub/Shrub	0.050	0.090	0.0700

Prepare Friction File

All 1 AOI(s) processed.
Per-AOI Friction outputs — click an AOI to preview its Manning map

AOI_02shp

LULC Manning preview

Land Cover Breakdown

Code	Type	Area (km²)	% area
2	Trees	278.780	71.1%
7	Built ...	71.870	18.3%
11	Range...	32.294	8.2%
5	Crops	6.912	1.8%
1	Water	2.076	0.5%
8	Bare ...	0.064	0.0%
4	Flood...	0.004	0.0%
10	Clouds	0.000	0.0%
	Total	392.000	100.0%

LULC Map

LULC — AOI_02shp

Manning's n Map

Manning n — AOI_02shp

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Step 5 — BC (Boundary Conditions)

Keep **Auto-detect from NHD (USA)**. The downstream boundary type defaults to **2 — Normal slope** with slope **0.001** — keep both:

5. Boundary Conditions (.src + .extbc)

Boundary geometry

Boundary coordinates: Auto-detect from NHD (USA) ↕

Downstream boundary condition

Boundary type: 0 — Free flow / supercritical outflow
1 — Water level vs time (upload stage file)
✓ 2 — Normal slope
3 — Froude number ↕

Normal slope: 0.00100 ▾

Click **Write .src + .extbc file(s)**.

Expected: the preview map shows Village Creek crossing the domain with the upstream inflow point (orange) on the northwest edge and the downstream boundary (red) at the southeast corner. Two files are generated and previewed: `<A0I>.src` (inflow point coordinates) and `<A0I>.extbc` (the type-2 outflow boundary segment on the DEM edge with slope 0.001).

FIMsim | Flood Inundation Model Simulation Tool v1.0 (TRITON)

1. Project2. AOI3. DEM4. Friction5. BC6. Hydrograph7. Config

mes directly.

1 AOI confirmed.

5. Boundary Conditions (.src + .extbc)

Boundary geometry

Boundary coordinates: Auto-detect from NHD (USA)

Downstream boundary condition

Boundary type: 2 — Normal slope

Normal slope: 0.00100

Write .src + .extbc file(s)

All 1 AOI(s) processed.

Per-AOI BC outputs — click an AOI to preview its files

AOI_02shpAOI_02shp.srcAOI_02shp.extbc

BC preview

AOI_02shp

Main river
Upstream
Downstream

.src (inflow points)

% X,Y
949565.956317,3379402.151203

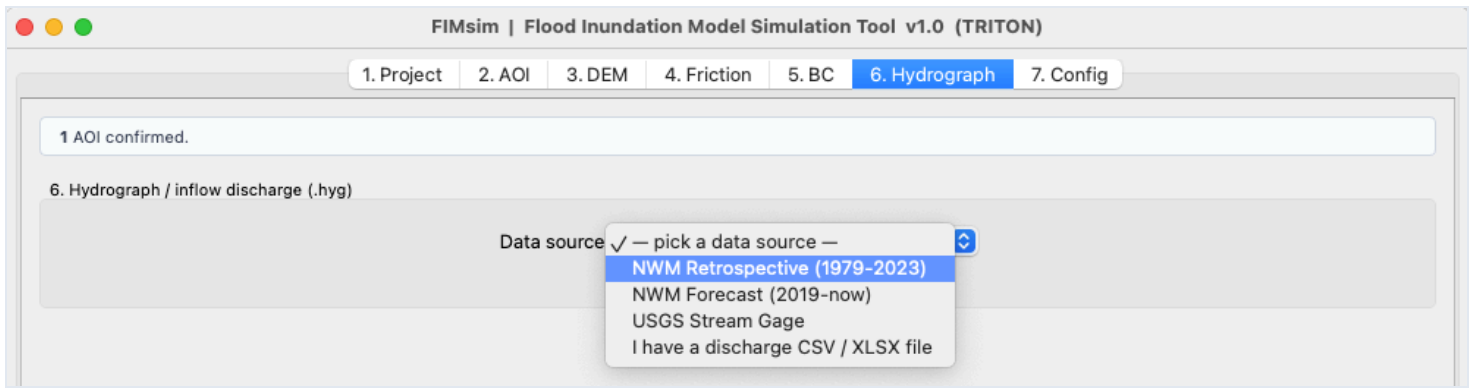
.extbc (outflow boundary)

% BC Type, X1, Y1, X2, Y2, BC
2,963495,3358658.024805,963495,3358997.10548,0.001

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Step 6 — Hydrograph

The hydrograph step offers the same data sources as LISFLOOD's BDY step — NWM Retrospective (1979–2023), NWM Forecast (2019–now), USGS Stream Gage, or your own CSV/XLSX file:



In Test Case 1 (LISFLOOD-FP) we used a **USGS stream gage**; this time we download the discharge from the **NWM** instead — simply to demonstrate both processes. Select **NWM Retrospective (1979–2023)** and keep **Feature ID: Auto-detect** (you can also enter a feature ID manually). Set:

- **Event start:** 2017-08-24 00:00
- **Event end:** 2017-09-10 00:00
- **Time interval:** 1.00 hours

Click **Write .hyg file(s)**.

Expected: the auto-detected feature ID is **1166365** (shown under the AOI name), and the hydrograph preview shows the Hurricane Harvey flood wave peaking near **3,250 m³/s around August 31**. The **<AOI>.hyg** file is written.

FIMsim | Flood Inundation Model Simulation Tool v1.0 (TRITON)

1. Project2. AOI3. DEM4. Friction5. BC6. Hydrograph7. Config

1 AOI confirmed.

6. Hydrograph / inflow discharge (.hyg)

Data source: NWM Retrospective (1979-2023)

Feature ID:

Auto-detect

Enter manually: e.g. 23212900

Event start: 2017-08-24 00:00

Event end: 2017-09-10 00:00

★ NWM v3.0 reanalysis, available 1979-02-01 → 2023-01-31 | resampled to chosen interval | USA only

Time interval: 1.00 hours

Write .hyg file(s)

All 1 AOI(s) processed.

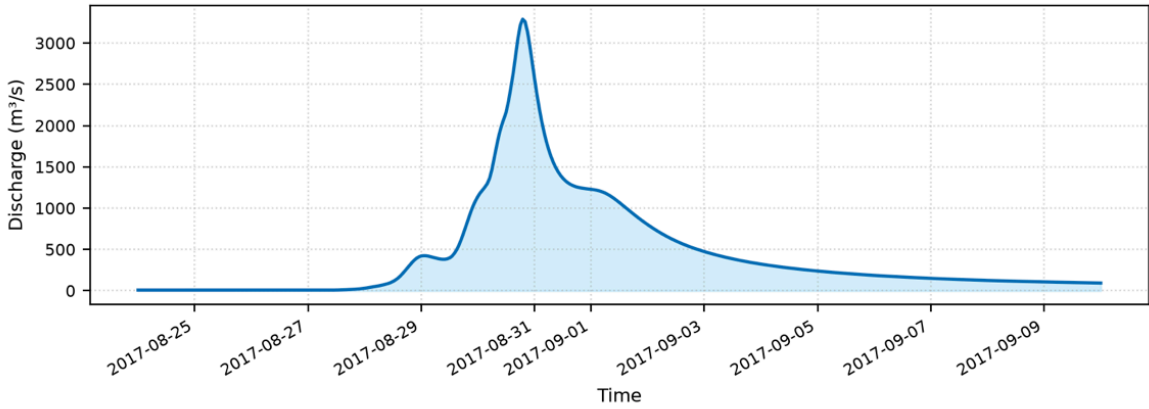
Per-AOI hydrographs — click an AOI to preview

AOI_02shp

NWM Retrospective. Feature ID: 1166365

Hydrograph preview

Hydrograph — AOI_02shp



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Step 7 — Config

The final step assembles the TRITON configuration file. Keep the defaults (ASC output format, time step, print interval — the simulation duration comes from the hydrograph automatically) and run.

Expected: <AOI>.cfg is written. The triton-files/ folder now contains the complete TRITON input package — dem.asc , friction.asc , <AOI>.src , <AOI>.extbc , <AOI>.hyg , <AOI>.cfg — ready to run with the TRITON solver.

FIMsim | Flood Inundation Model Simulation Tool v1.0 (TRITON)

1. Project2. AOI3. DEM4. Friction5. BC6. Hydrograph7. Config

★ The TRITON .cfg is generated automatically for each AOI from the prepared DEM / friction / BC / hydrograph. sim_duration is taken from the hydrograph length. Adjust the few options below if needed.

1 AOI confirmed.

7. TRITON config (.cfg)

Config options (the rest is auto-generated)

Output format:ASC

Print option:huv

Time step:10.000 s

Print interval:3600.00 s

Courant number:0.50

Generate .cfg file(s)

All 1 AOI(s) processed.

Per-AOI .cfg — click an AOI to preview the generated file

AOI_02shpAOI_02shp.cfg

.cfg preview

#-----
AOI_02shp
TRITON parameter file generated by FIMsim.
#-----
Input Data
#-----
Topography
dem_filename="dem.asc"

Manning / friction input
n_infile="friction.asc"
#const_mann=0.035

Hydrograph and source locations

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Troubleshooting

- **A download step fails or times out** — the USGS / NOAA data services occasionally have outages; simply re-run the step.
- **NWM Retrospective date limits** — the archive covers **Feb 1979 – Jan 2023**; the NWM Forecast archive starts in 2019.
- **USGS gage data** — retrieved from waterdata.usgs.gov; 15-minute readings are resampled to your chosen time interval.